



LONGEVITY & ANTI-AGING

NAD+

NAD+ – 500 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	beta-Nicotinamide adenine dinucleotide
SYNONYMS / ALIASES	NAD+; NAD (oxidized form)
CAS NUMBER	53-84-9
MOLECULAR FORMULA	C21H27N7O14P2
MOLECULAR WEIGHT	663.43 g/mol
AMINO ACID SEQUENCE	N/A (not a peptide)
SOURCE	Synthetic

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	500 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Sensitive to heat and light in solution.

STRUCTURAL NOTES

NAD⁺ is a pyridine dinucleotide coenzyme composed of nicotinamide mononucleotide and adenosine monophosphate joined by a pyrophosphate bridge. It is a small-molecule dinucleotide and not a peptide. This entry provides a descriptive chemical classification only.

SELECTED PEER-REVIEWED REFERENCES

1. Belenky P. et al. - *NAD⁺ metabolism in health and disease* - Trends Biochem Sci, 2007 - DOI: 10.1016/j.tibs.2006.11.006
2. Verdin E. - *NAD⁺ in aging, metabolism, and neurodegeneration* - Science, 2015 - DOI: 10.1126/science.aac4854

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.